Workforce & Retention

A guide to predicting and retaining your future

Analysis by DDSAnalytics

Agenda

- Attrition Summary
- Data Overview
- Feature Selection
- Preliminary Models

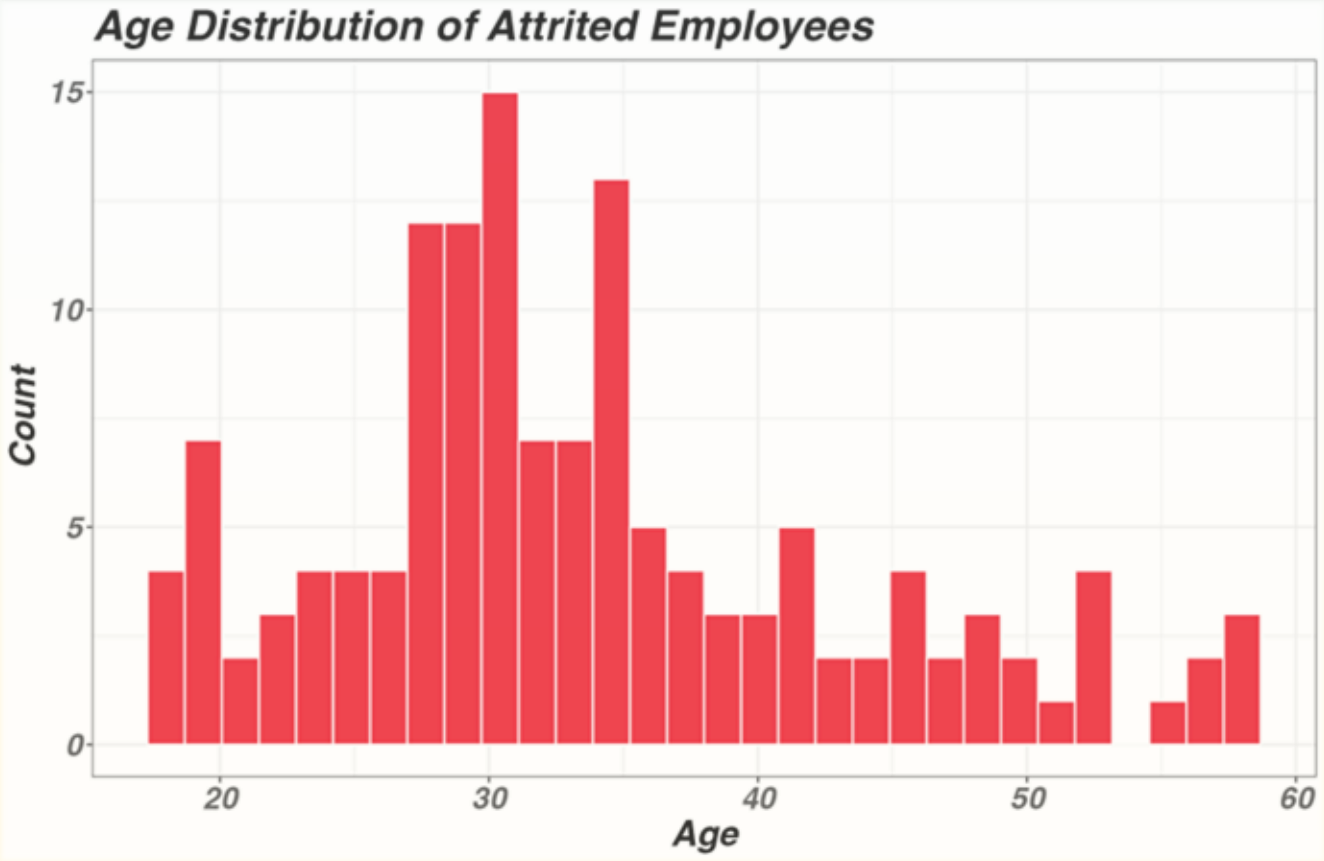
Attrition Summary

- Mercer's Annual US Turnover Survey:

  ~ 13%

- FritoLays:

  16.1%



Reported 870 employees (including 140 attritted)

Attrition Summary

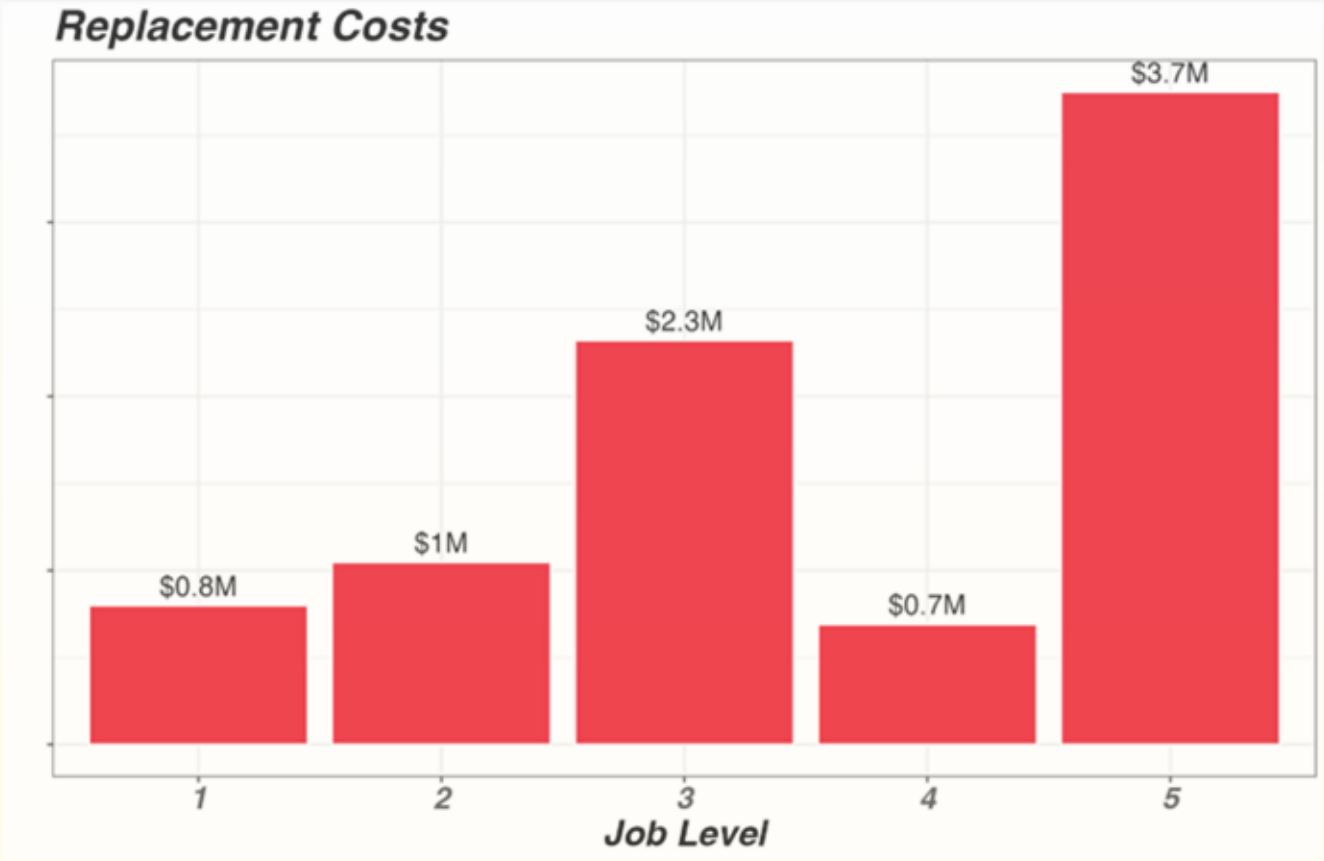
- Mercer's Annual US Turnover Survey:

  ~ 13%

- FritoLays:

  16.1%

- $8.5 million in replacement costs
- Costs realized within 6-9 months
- 3% reduction = $1.6 million saved
- 6% reduction = $3.2 million saved



Attrition Summary

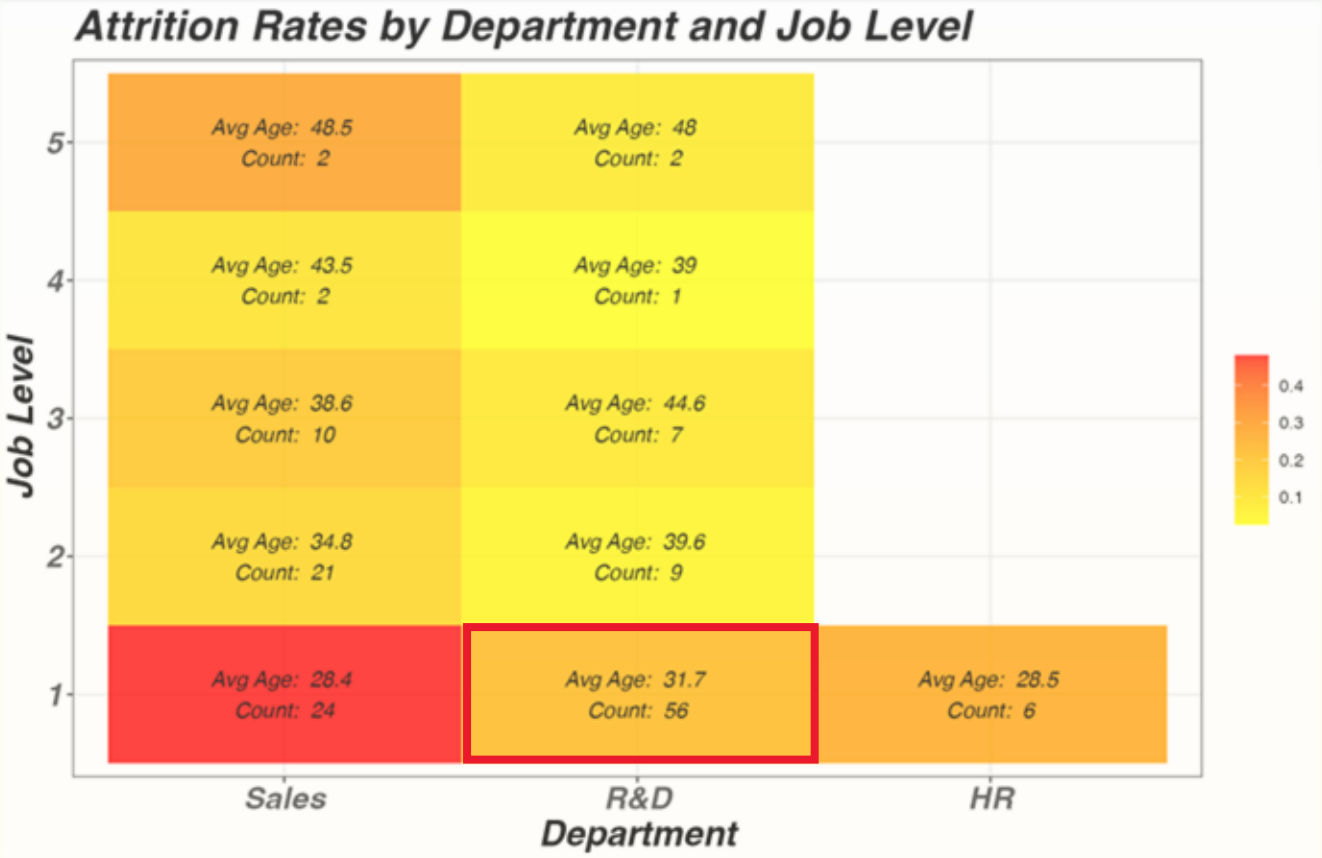
- Mercer's Annual US Turnover Survey:

  ~ 13%

- FritoLays:

  16.1%

- $8.5 million in replacement costs
- Costs realized within 6-9 months
- 3% reduction = $1.6 million saved
- 6% reduction = $3.2 million saved
- R&D presents largest opportunity
- Sales focus retaining level 2 and above

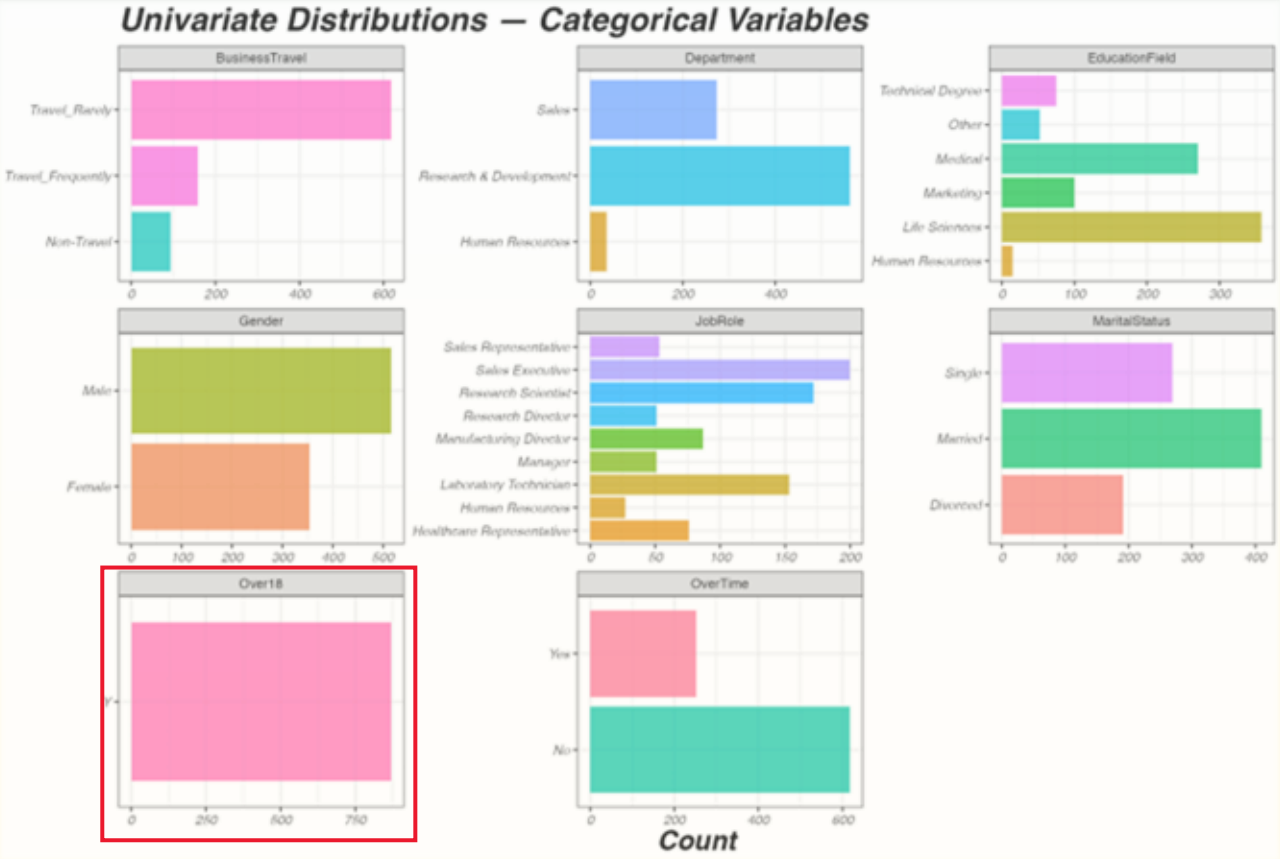


Data Overview

- Constants & Sequentials:
  - Standard Hours, Employee Count, Employee ID

Data Overview

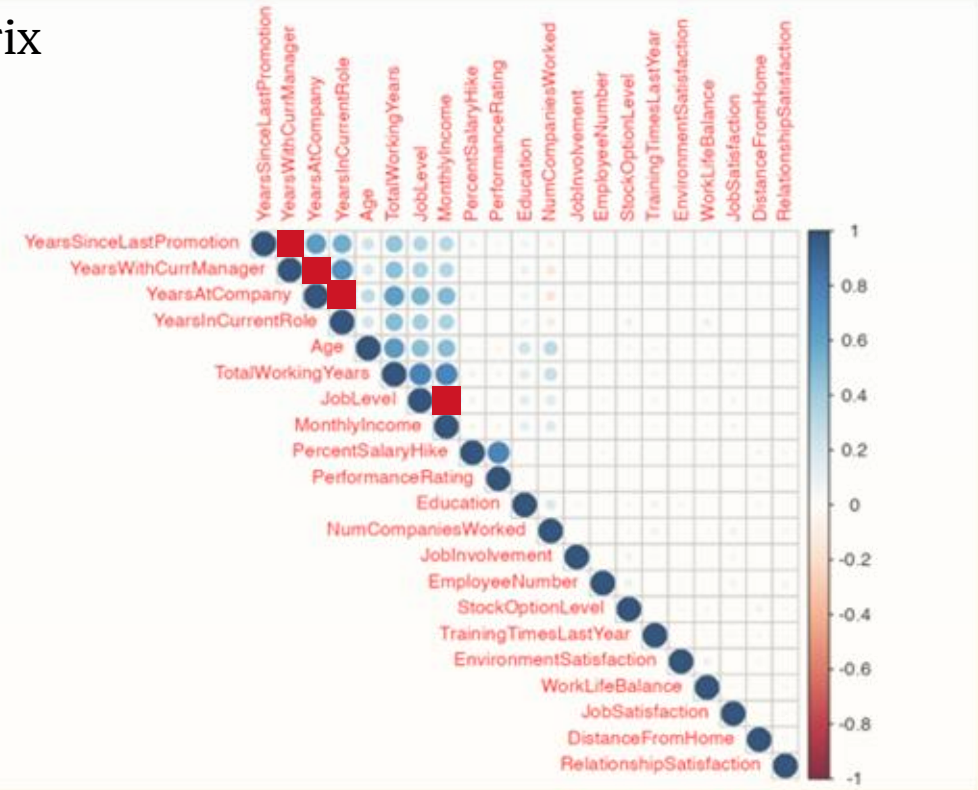
- Constants & Sequentials:
  - Standard Hours, Employee Count, Employee ID
  - Over 18



Data Overview

- Constants & Sequentials:
  - Standard Hours, Employee Count, Employee ID
  - Over 18
- Redundant Pairs:
  - Monthly Income, Job level
  - Years at Company, Years in Current Role, Years with Current Manager, Years since last Promotion

Pairs Matrix



Data Overview

- Constants & Sequentials:
  - Standard Hours, Employee Count, Employee ID
  - Over 18
- Redundant Pairs:
  - Monthly Income, Job level
  - Years at Company, Years in Current Role, Years with Current Manager, Years since last Promotion

Pairs Matrix

JobLevel	95%	MonthlyIncome
JobLevel	78%	TotalWorkingYears
MonthlyIncome	78%	TotalWorkingYears
YearsAtCompany	78%	YearsInCurrentRole
PercentSalaryHike	78%	PerformanceRating
YearsAtCompany	77%	YearsWithCurrManager
YearsInCurrentRole	71%	YearsWithCurrManager
Age	65%	TotalWorkingYears
YearsAtCompany	64%	YearsSinceLastPromotion
YearsAtCompany	64%	TotalWorkingYears
YearsInCurrentRole	55%	YearsSinceLastPromotion
YearsAtCompany	52%	JobLevel
YearsSinceLastPromotion	51%	YearsWithCurrManager

Data Overview

- Constants & Sequentials:
  - Standard Hours, Employee Count, Employee ID
  - Over 18
- Redundant Pairs:
  - Monthly Income, Job level
  - Years at Company, Years in Current Role, Years with Current Manager, Years since last Promotion
  - Environment Satisfaction, Job Satisfaction, Relationship Satisfaction, Work Life Balance

Data Overview

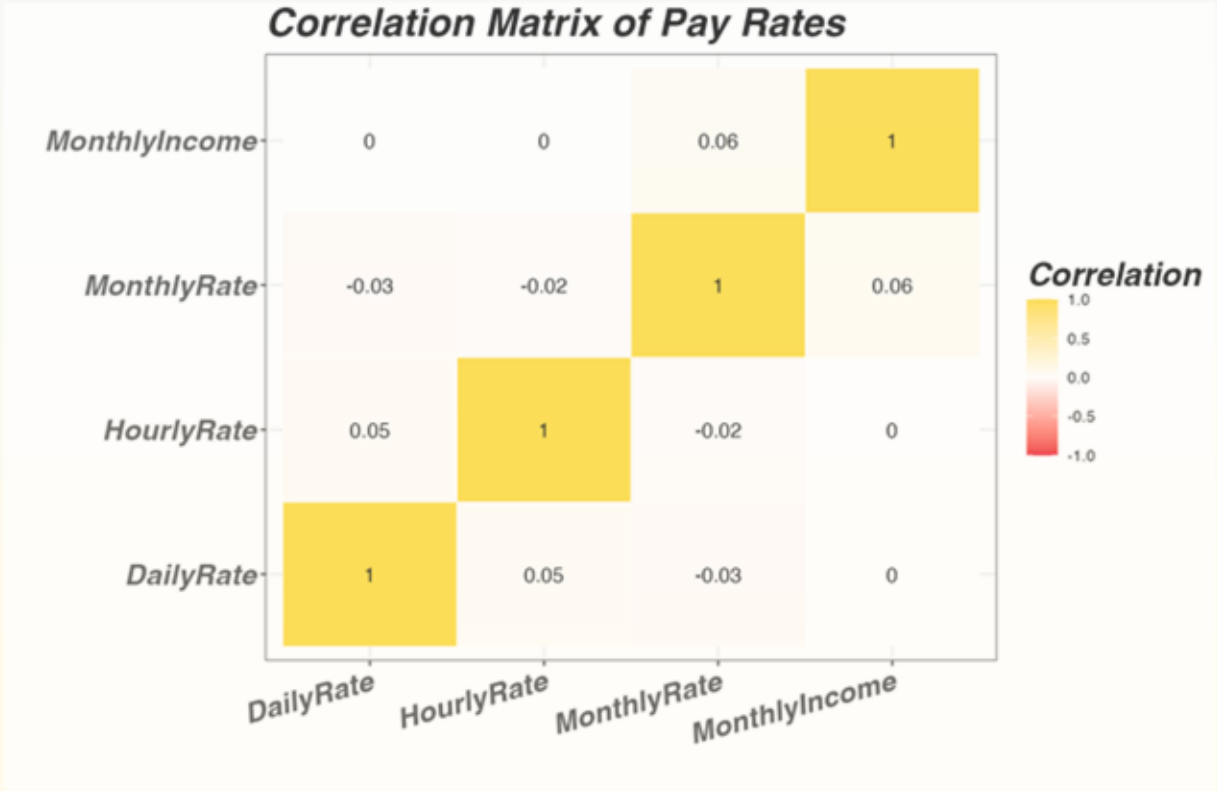
- Constants & Sequentials:
  - Standard Hours, Employee Count, Employee ID
  - Over 18
- Redundant Pairs:
  - Monthly Income, Job level
  - Years at Company, Years in Current Role, Years with Current Manager, Years since last Promotion
  - Environment Satisfaction, Job Satisfaction, Relationship Satisfaction, Work Life Balance
  - Monthly Income, Hourly Rate, Daily Rate, Monthly Rate

Data Overview

- Constants & Sequentials:
  - Standard Hours, Employee Count, Employee ID
  - Over 18
- Redundant Pairs:
  - Monthly Income, Job level
  - Years at Company, Years in Current Role, Years with Current Manager, Years since last Promotion
  - Environment Satisfaction, Job Satisfaction, Relationship Satisfaction, Work Life Balance
  - Monthly Income, Hourly Rate, Daily Rate, Monthly Rate

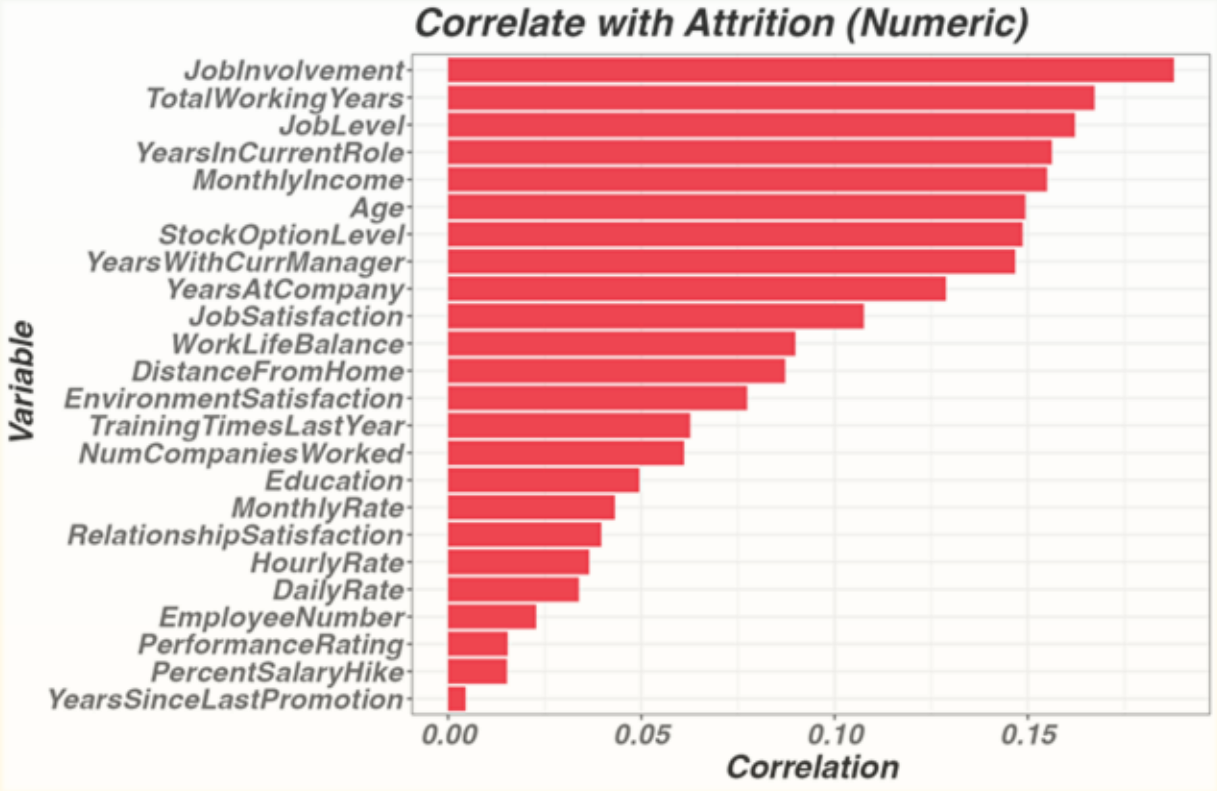
Data Overview

- Constants & Sequentials:
  - Standard Hours, Employee Count, Employee ID
  - Over 18
- Redundant Pairs:
  - Monthly Income, Job level
  - Years at Company, Years in Current Role, Years with Current Manager, Years since last Promotion
  - Environment Satisfaction, Job Satisfaction, Relationship Satisfaction, Work Life Balance
  - Monthly Income, Hourly Rate, Daily Rate, Monthly Rate



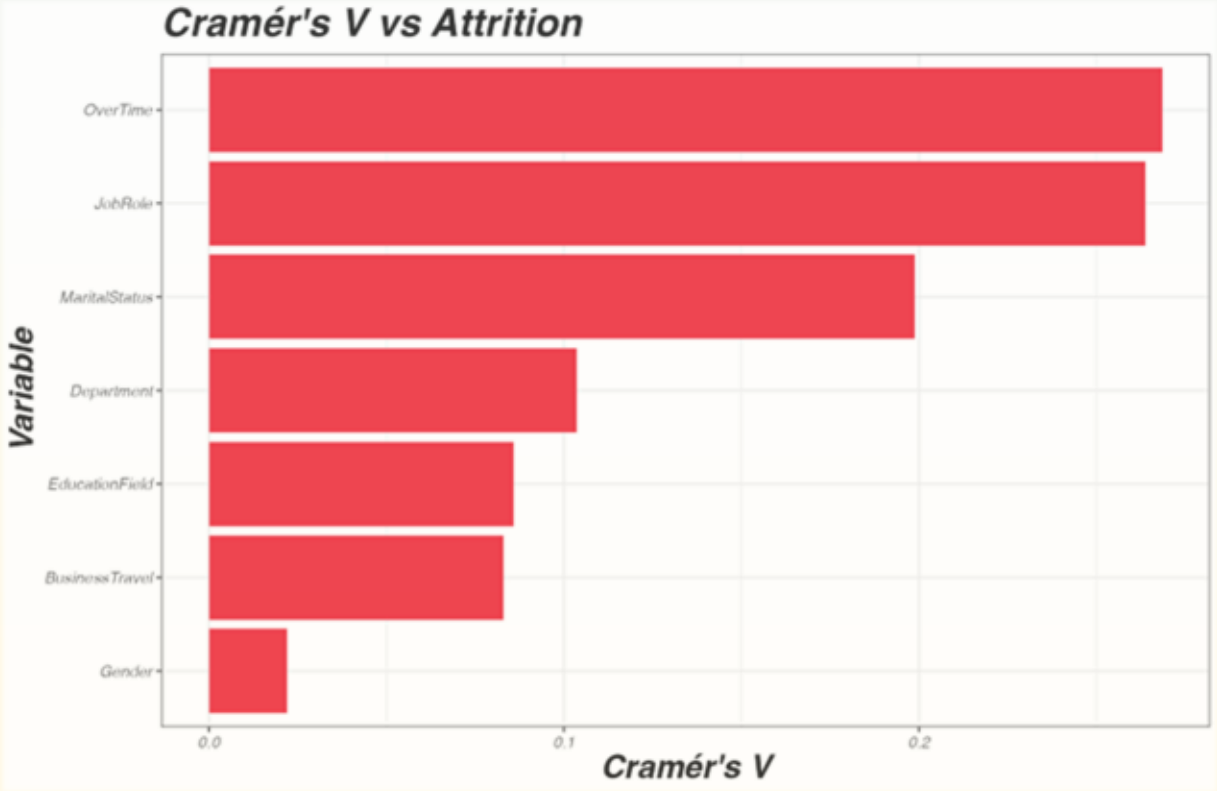
Data Overview

- Constants & Sequentials:
  - Standard Hours, Employee Count, Employee ID
  - Over 18
- Redundant Pairs:
  - Monthly Income, Job level
  - Years at Company, Years in Current Role, Years with Current Manager, Years since last Promotion
  - Environment Satisfaction, Job Satisfaction, Relationship Satisfaction, Work Life Balance
  - Monthly Income, Hourly Rate, Daily Rate, Monthly Rate,
- High Attrition Correlation:
  - Job Involvement, Total Working Years, Job Level, Years in Current Role, Monthly Income



Data Overview

- Constants & Sequentials:
  - Standard Hours, Employee Count, Employee ID
  - Over 18
- Redundant Pairs:
  - Monthly Income, Job level
  - Years at Company, Years in Current Role, Years with Current Manager, Years since last Promotion
  - Environment Satisfaction, Job Satisfaction, Relationship Satisfaction, Work Life Balance
  - Monthly Income, Hourly Rate, Daily Rate, Monthly Rate,
- High Attrition Correlation:
  - Job Involvement, Total Working Years, Job Level, Years in Current Role, Monthly Income
  - Overtime, Job Role, Marital Status



Cramer's V: Strength of association (Strong > 40%, Moderate > 20%, Weak < 20%)

Feature Selection

Retained

- Over Time
- Job Role
- Job Level
- Monthly Income
- Years with Current Manager
- Distance From Home
- Marital Status
- Age
- Stock Option Level
- Department
- Job Involvement
- Number of Companies Worked

Excluded

- Standard Hours
- Employee Count
- Over 18
- Employee ID
- Hourly Rate
- Daily Rate
- Monthly Rate
- Gender
- Years at Company
- Years in Current Role
- Years Since Last Promotion
- Percent Salary Hike
- Performance Rating
- Education
- Education field
- Business Travel

Engineered

- Total Satisfaction:
  - Environment Satisfaction
  - Job Satisfaction
  - Relationship Satisfaction
  - Work Life Balance

Ordered by supporting evidence

Preliminary Models

Environment

- All variables except:
  - Standard Hours
  - Employee Count
  - Over 18
  - Employee ID
- 70/30 Training to Test Ratio
  - Equivalent data used to train and test
- Reproducible instance
  - Eliminates random chance

K Nearest Neighbor

Accuracy 86%		Actual N	Actual Y
Predicted	N	219	35
	Y	0	7
Misclassified 14%		Specificity 100%	Sensitivity 17%

- Uses average of 5 nearest points
- Future tuning will determine best K

Naïve Bayes

Accuracy 75%		Actual N	Actual Y
Predicted	N	172	19
	Y	47	23
Misclassified 25%		Specificity 79%	Sensitivity 55%

- May provide better sensitivity at the cost of incorrectly identifying potential employees at risk of attrition

Next steps: Tuning with iterations and Optimizing with feature selection

Workforce & Retention

Questions